CALIFORNIA HOUSING

Comprehensive Analysis

github.com/Daniyaliranmehr/Document-Question-Answering-System

PROJECT OVERVIEW

This project presents a comprehensive data science and deep learning analysis of the California Housing dataset. The primary objective is to predict median house values using a fully connected neural network implemented from scratch in PyTorch while systematically investigating how different data preparation techniques, loss functions, and optimization algorithms affect model performance.

This project was developed for educational and research purposes and serves as a practical exploration of data science, neural network regression, robust loss functions, and optimization techniques for real-world tabular datasets.

PROJECT PHASES

1

DATA SCIENCE

Data analysis and preparation

2

LOSS FUNCTION ANALYSIS

Comparison and analysis of regression loss functions

3

OPTIMIZER ANALYSIS

Comparison and analysis of optimization algorithms

4

FINAL TESTING

Testing the best-performing model on unseen test data.

1 | DATA SCIENCE

DATA UNDERSTANDING

- Dataset structure
- Initial Statistical Insights
- Missing Values

EXPLORATORY DATA ANALYSIS (EDA)

- Univariate Analysis
- Correlation Analysis
- Bivariate Analysis
- Geographic Visualization

DATA PREPARATION

- Data Cleaning
- Feature Engineering
- Feature Transformation
- Encoding
- Feature Scaling

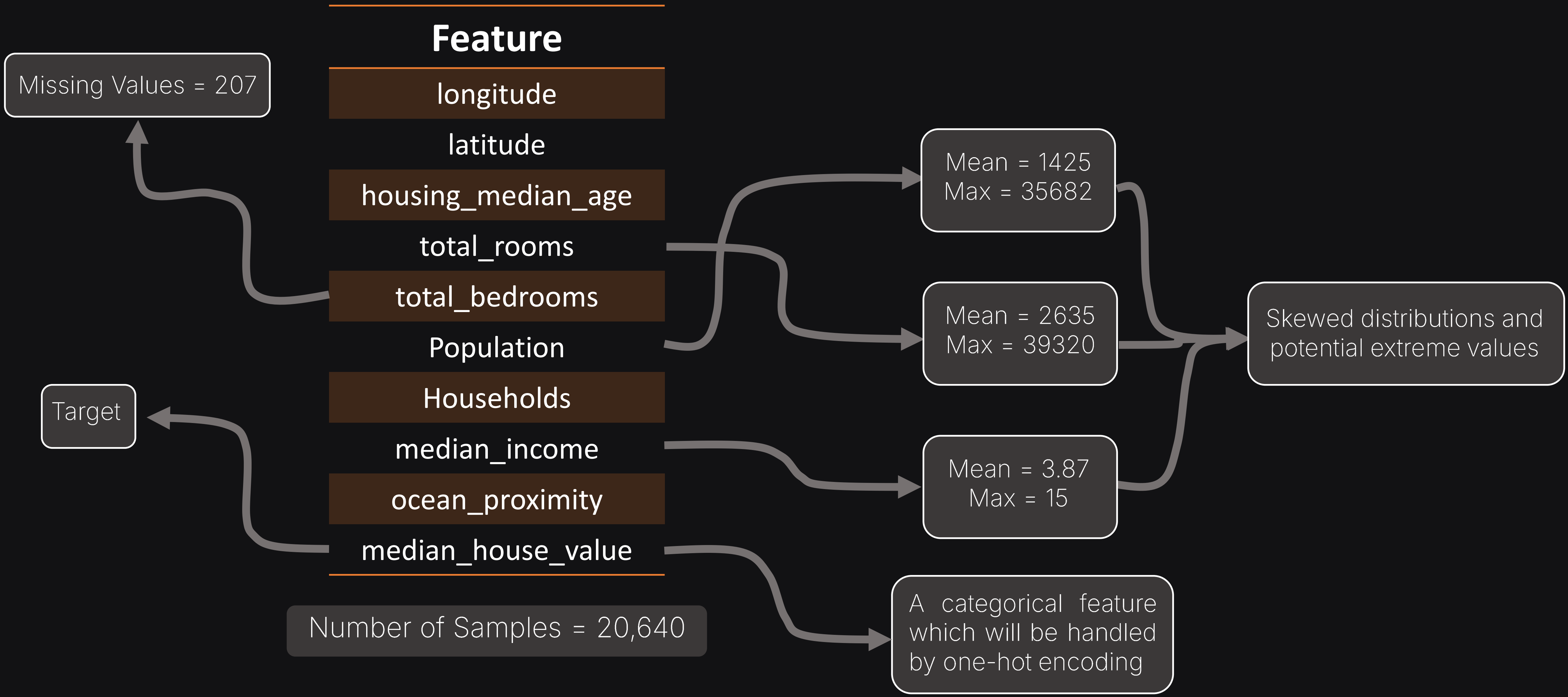
1 | DATA SCIENCE

The Data Science phase was a crucial part of this project. I found that some features were capped, most features were significantly skewed, the dataset contained a considerable number of outliers, and missing values were present. I applied preprocessing techniques such as outlier treatment, feature engineering, feature transformation, and data standardization to improve data quality and model readiness.

This process not only contributed to better model performance but also strengthened my understanding of how data characteristics influence the behavior of deep learning models. More importantly, it demonstrated that successful predictive modeling depends not only on model architecture but also on careful data preparation and analysis.

1 | DATA SCIENCE

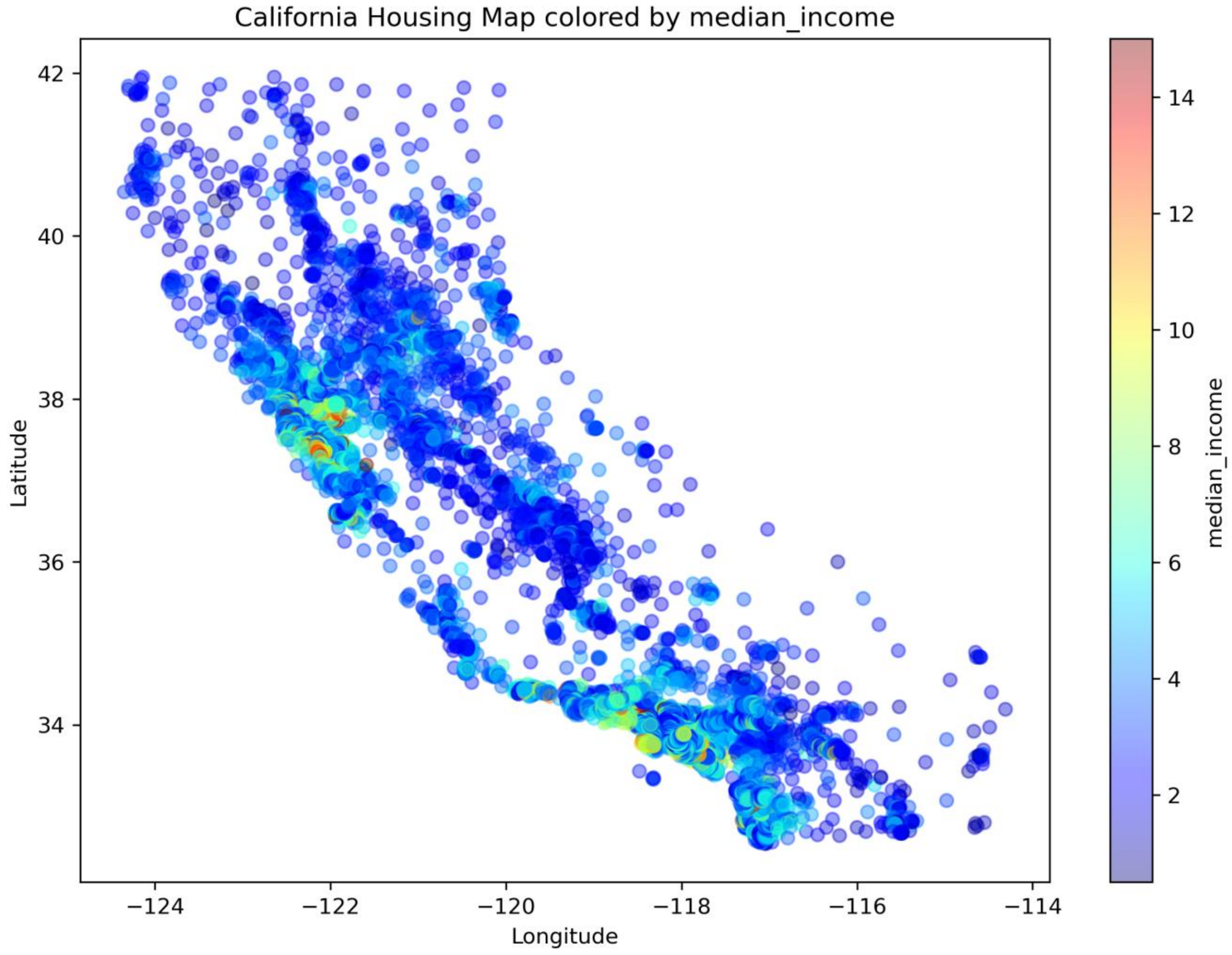
DATA UNDERSTANDING



1 | DATA SCIENCE

EXPLORATORY DATA ANALYSIS

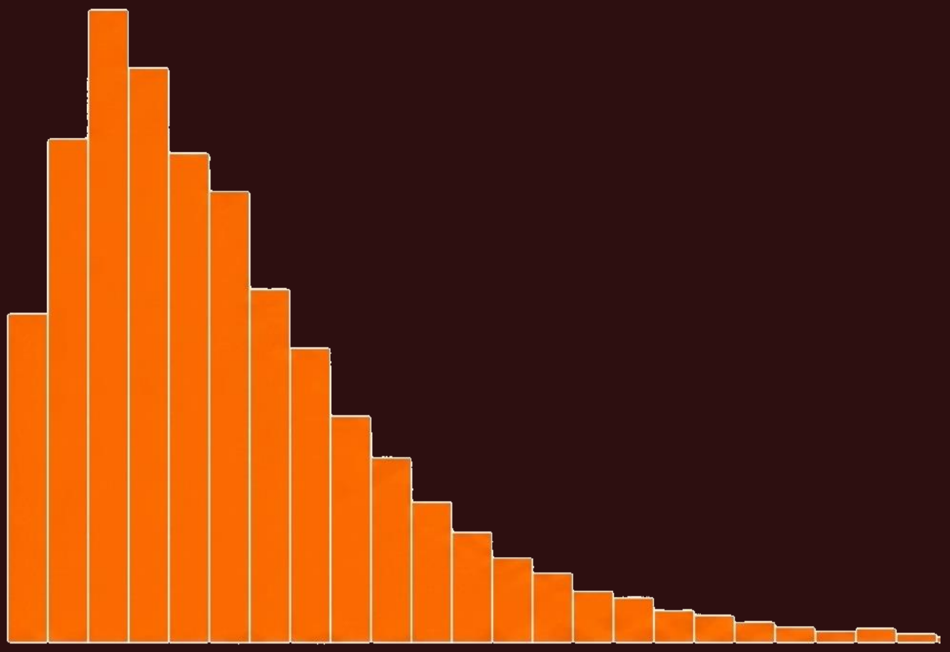
Through correlation analysis, bivariate analysis, and geographic visualization, I examined how each feature contributes to the target variable. The analysis revealed that **median income** is the most important predictor of house value, exhibiting the strongest relationship with the target among all features.



1 | DATA SCIENCE

EXPLORATORY DATA ANALYSIS

Using histograms, boxplots, and IQR analysis, I examined the distribution of features and validated initial assumptions about the dataset. This analysis revealed significant **skewness** in most features and identified a considerable number of **outliers**, highlighting the need for feature transformation and outlier treatment before model training.



IQR

Interquartile Ranga

Lower Bound = Q1 − 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR

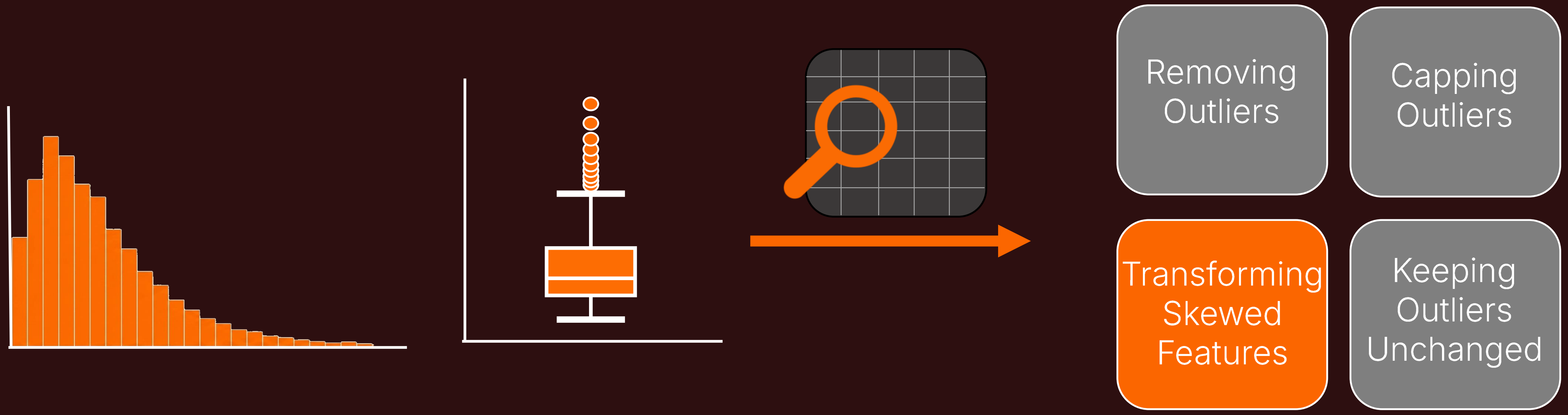
1 | DATA SCIENCE
DATA PREPARATION

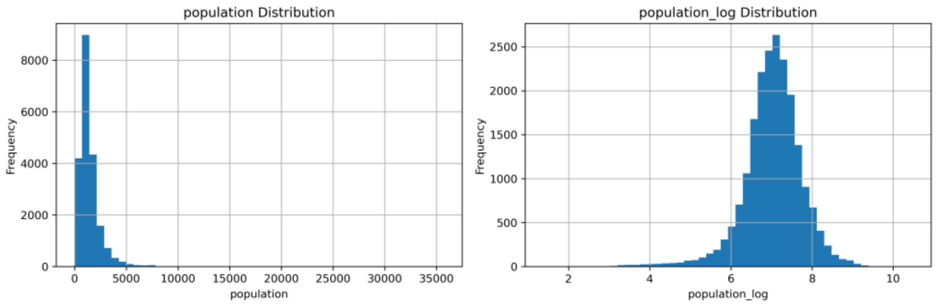
The total_bedrooms feature contained 207 missing values (≈1% of the dataset). After evaluating several imputation strategies, median imputation was selected because the feature is right-skewed and contains outliers. Compared to the mean, the median provides a more robust and representative estimate while preserving all observations in the dataset.

1 | DATA SCIENCE

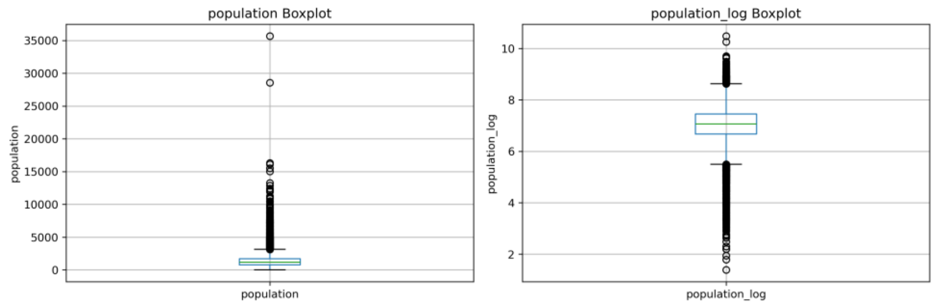
DATA PREPARATION

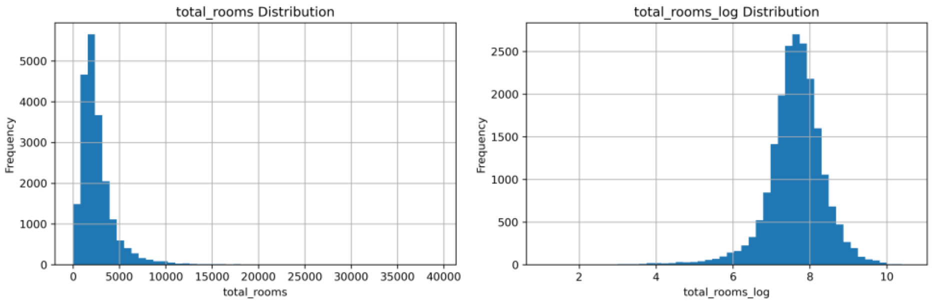
Several features contained extreme values, but the analysis indicated that these observations were valid and informative rather than erroneous. To avoid losing valuable information, the outliers were retained. Their influence was later reduced through feature transformation, which helped address skewness while preserving the original observations.



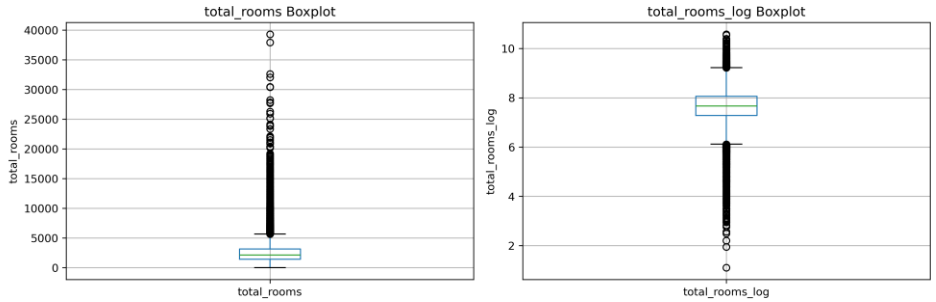


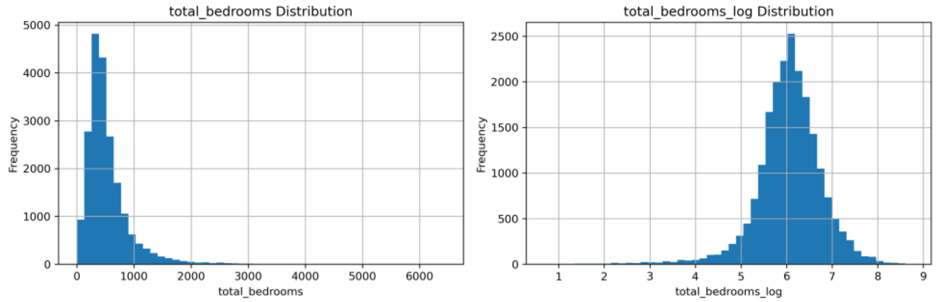
population



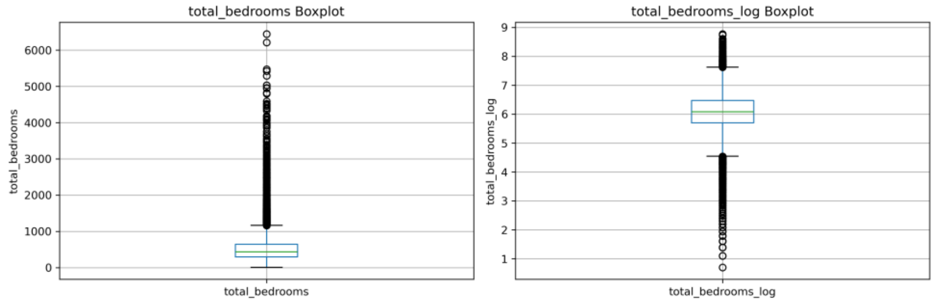


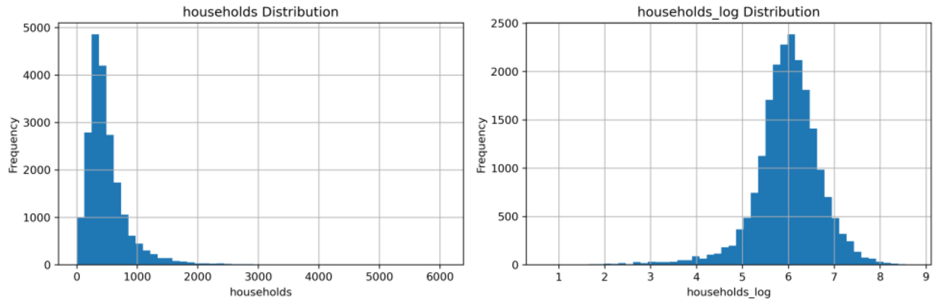
total_rooms



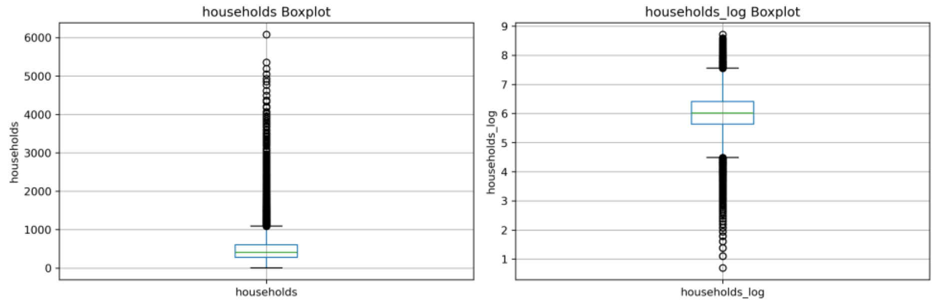


total_bedrooms





households



1 | DATA SCIENCE

DATA PREPARATION

The original features mainly represented district size and did not directly capture living conditions or neighborhood characteristics. To provide more meaningful information to the model, I engineered three ratio-based features: rooms_per_household, bedrooms_per_room, and population_per_household.

The analysis showed that these engineered features contain more useful predictive information than several of the original count-based features. In particular, bedrooms_per_room exhibited the strongest relationship with house value, indicating that housing composition is an important factor in price prediction.

Visual analysis also revealed skewed distributions, extreme values, and a small number of unrealistic observations. To improve data quality and prepare the features for neural network training, appropriate capping and feature transformations were applied while preserving their underlying information.

Correlation Analysis of Engineered Features

Feature	Correlation
total_rooms	0.134153
total_bedrooms	0.049457
bedrooms_per_room	-0.245496

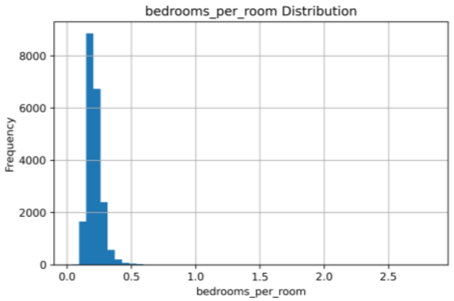
Even prior to transformation, this engineered ratio uncovers the strongest negative linear relationship with the target variable among all density features. The clipping process maintains this strong predictive signal.

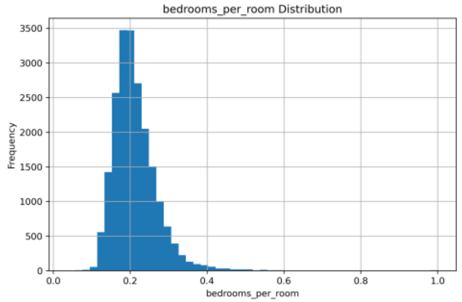
Feature	Correlation
total_rooms	0.134153
households	0.065843
rooms_per_household	0.260288

Normalizing the raw room count by household units, followed by mathematical scaling, provides a cleaner and more stable linear connection to the target variable.

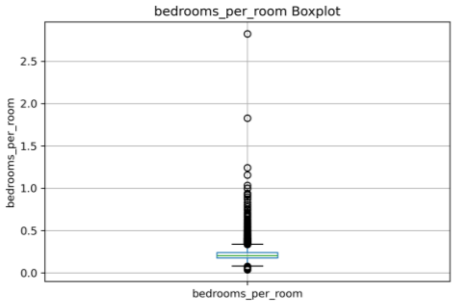
Feature	Correlation
population	-0.024650
households	0.065843
population_per_household	-0.277509

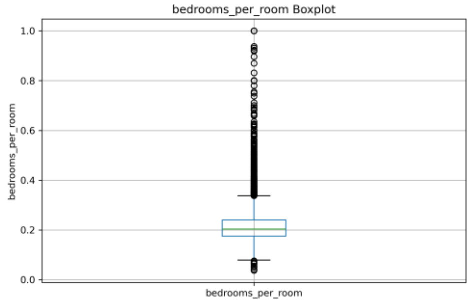
Normalizing absolute population counts into a household density ratio, followed by range compression, uncovers a massively amplified predictive signal for the target variable.

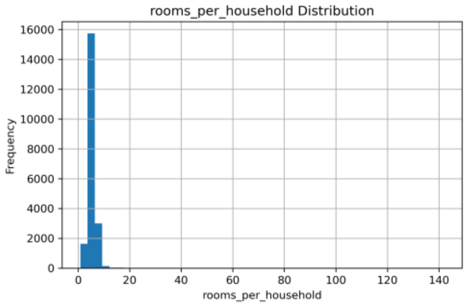


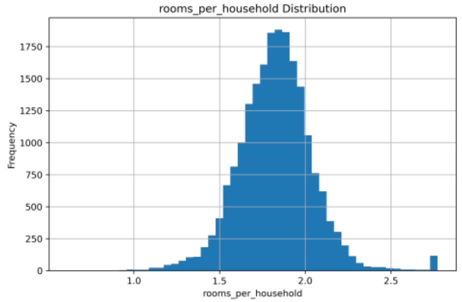


bedrooms_per_room

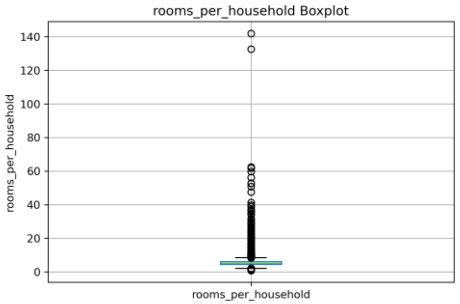


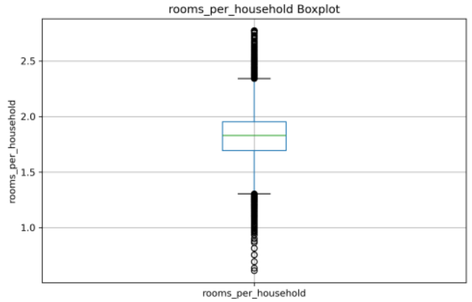


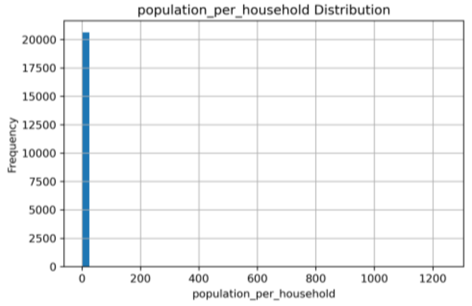


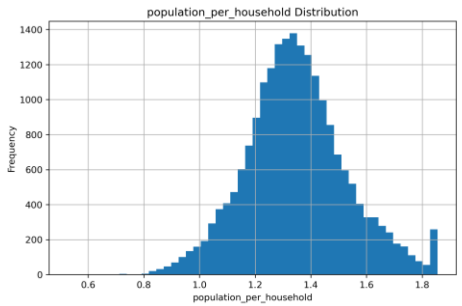


rooms_per_household

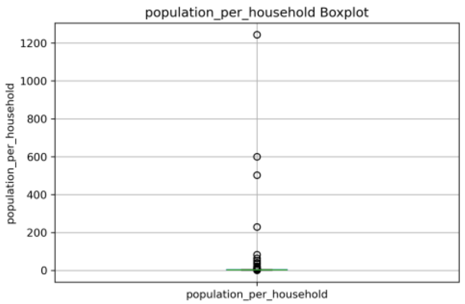


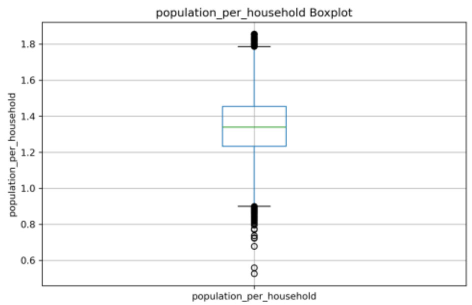






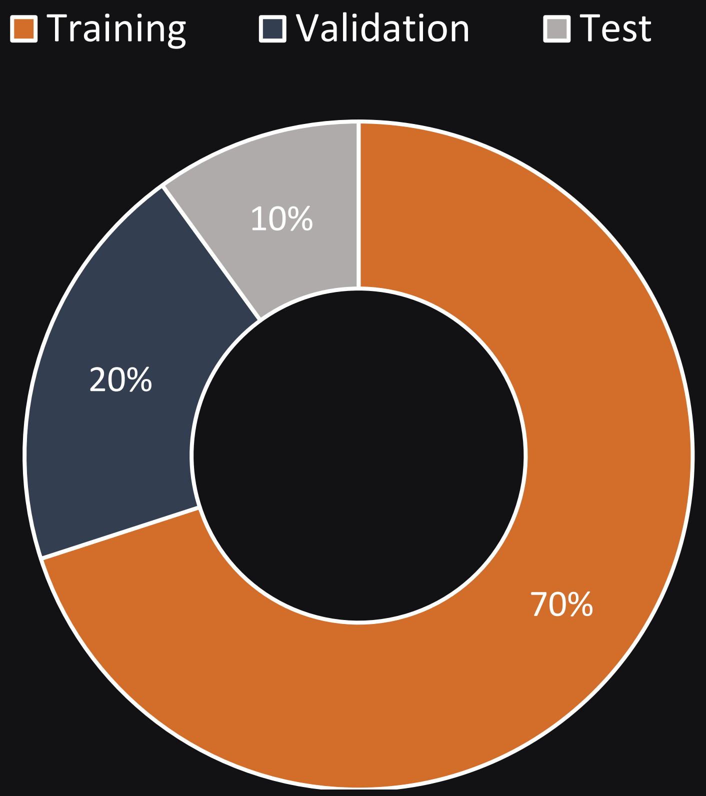
population_per_household





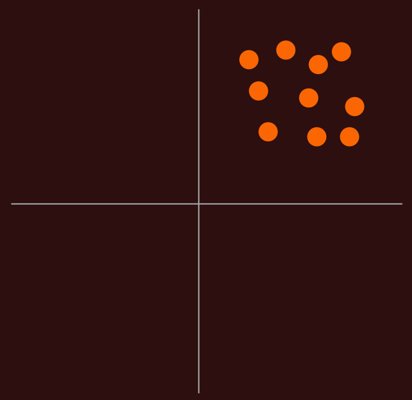
1 | DATA SCIENCE
DATA PREPARATION

The dataset is split into training, validation, and test sets.



1 | DATA SCIENCE
DATA PREPARATION

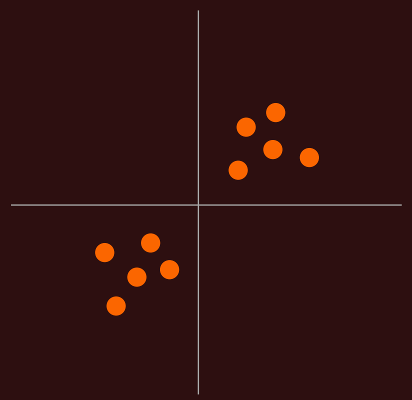
Feature scaling is applied to ensure that all numerical features contribute equally during model training. Since different features in the dataset have varying ranges and magnitudes, models (especially gradient-based methods) may become biased toward features with larger values. Standardization (Z-score normalization helps improve training stability and convergence speed by transforming features into a common scale.



$$z = \frac{x - \mu}{\sigma}$$

Where:

- μ is mean
- σ is standard deviation



DEEP LEARNING

I analyzed the training and validation loss and R^2 learning curves over 300 epochs for each loss function. The learning curves were used to evaluate convergence behavior, detect potential overfitting or underfitting, and identify the best-performing checkpoint, which was saved during training.

In the loss function analysis section, the optimizer was fixed to Adam, whereas in the optimizer analysis section, MAE was used as the loss function. All models were trained with a learning rate of 0.001 to ensure a fair comparison.

2 | LOSS FUNCTION ANALYSIS

MSE

$$\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

MAE

$$\frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

Huber

$$\begin{cases} \frac{1}{2}e^2, & |e| \le \delta \\ \delta\left(|e| - \frac{1}{2}\delta\right), & |e| > \delta \end{cases}$$

Log-Cosh

$$\frac{1}{N}\sum_{i=1}^{N}\log(\cosh(\hat{y}_i - y_i))$$

Quadratic

$$\frac{1}{2}\left(\frac{x}{c}\right)^2$$

Cauchy

$$\log\left(\frac{1}{2}\left(\frac{x}{c}\right)^2 + 1\right)$$

Welsch

$$1 - \exp\left(-\frac{1}{2}\left(\frac{x}{c}\right)^2\right)$$

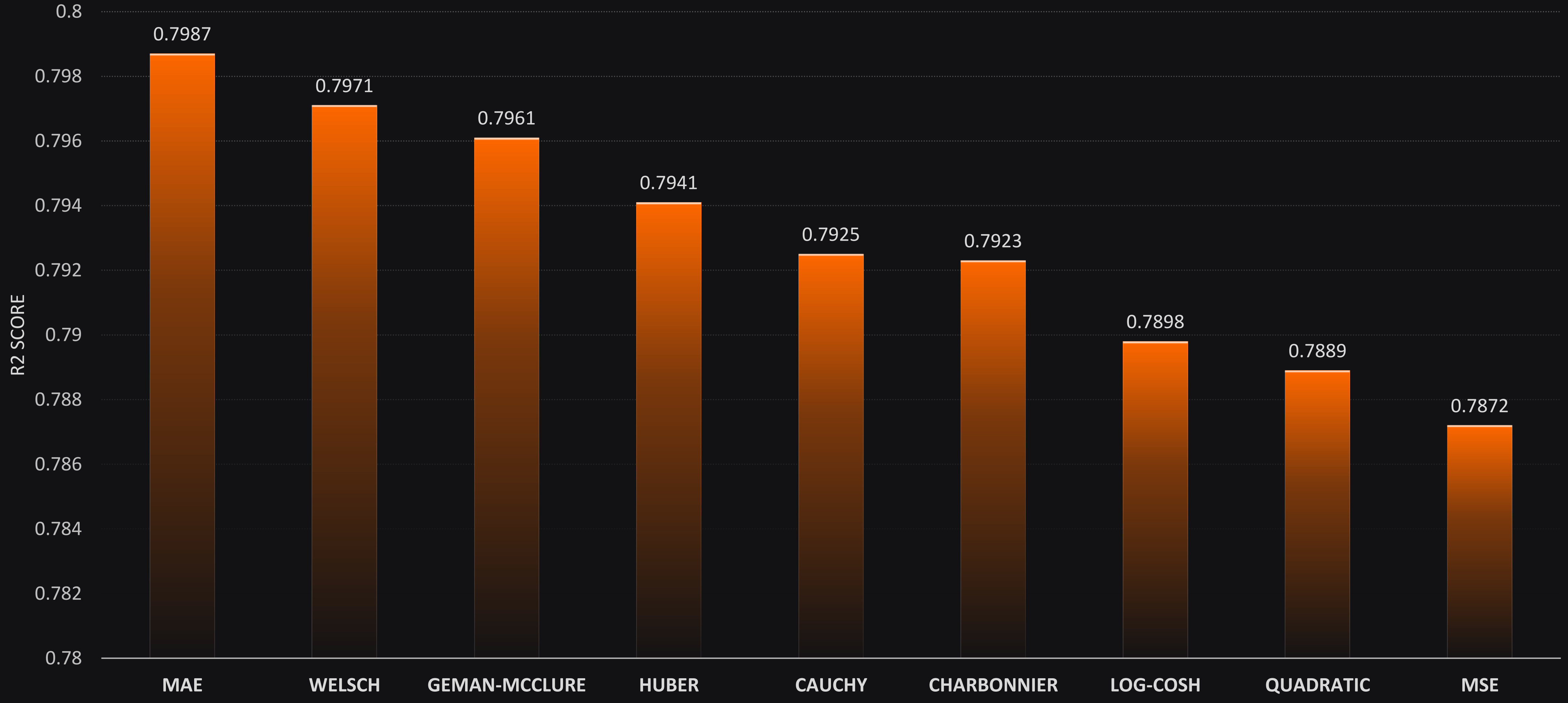
Charbonnier

$$\frac{|\alpha - 2|}{\alpha}\left(\left(\frac{\left(\frac{x}{c}\right)^2}{|\alpha - 2|} + 1\right)^{\frac{\alpha}{2}} - 1\right) \; ; (\alpha = 1)$$

Geman-McClure

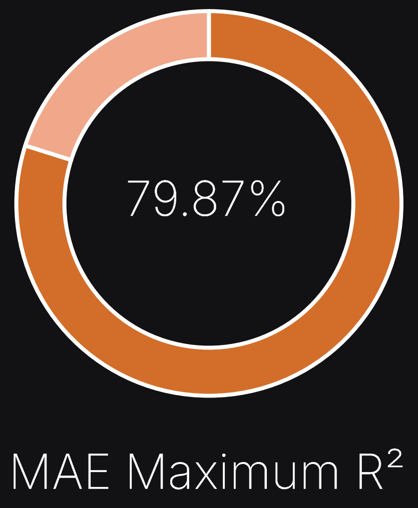
$$\frac{|\alpha - 2|}{\alpha}\left(\left(\frac{\left(\frac{x}{c}\right)^2}{|\alpha - 2|} + 1\right)^{\frac{\alpha}{2}} - 1\right) \; ; (\alpha = -2)$$

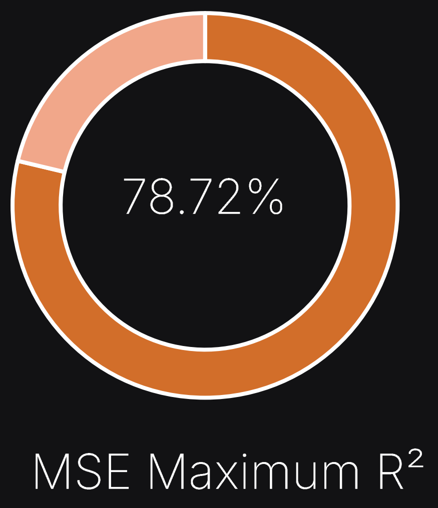
2 | LOSS FUNCTION RESULTS

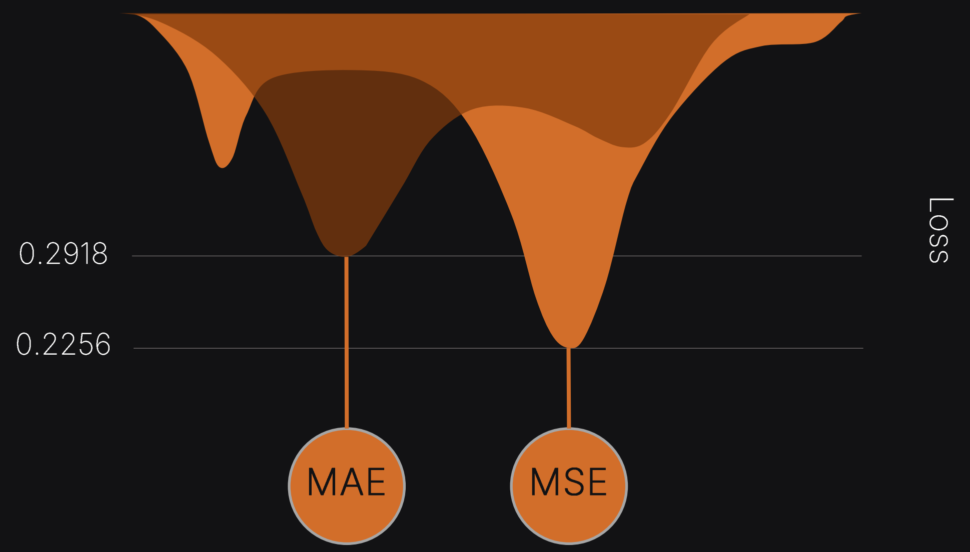


2 | LOSS FUNCTION INSIGHTS

The MAE loss function achieved the highest R^2 score (0.7987), while MSE achieved the lowest (0.7872). The difference was only 0.0115, indicating that the choice of loss function had a limited impact on overall performance. Although MSE produced a lower loss value (0.2256) than MAE (0.2918), these values are not directly comparable because they are computed using different objective functions. Based on the common evaluation metric (R^2), MAE delivered the best predictive performance.







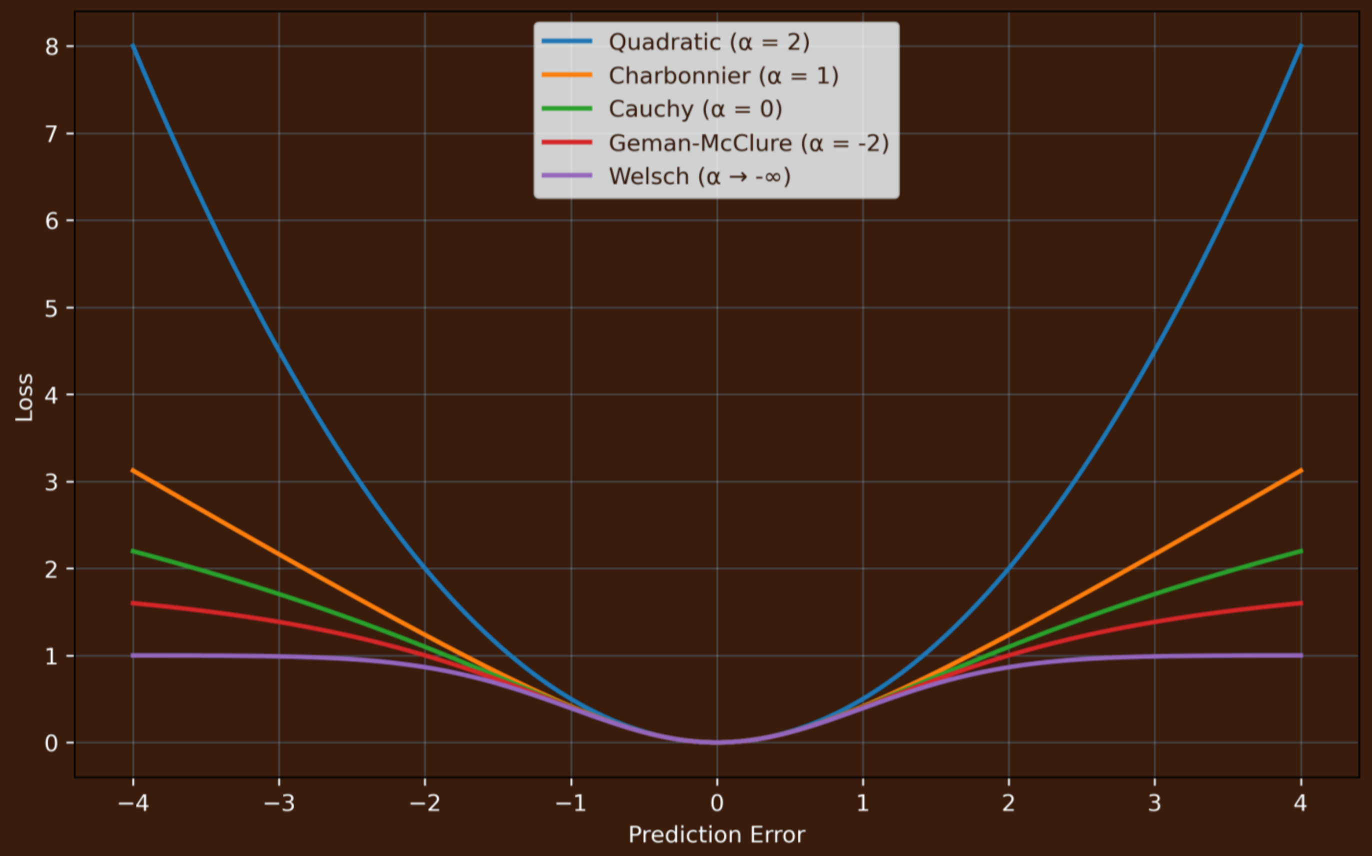
Training History MAE
Train Loss
Validation Loss
Train R²
Validation R²
Loss
R²
Epoch
0.50
0.45
0.40
0.35
0.30
0.25
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0
50
100
150
200
250
300

2 | LOSS FUNCTION INSIGHTS

The California Housing dataset is not perfectly normally distributed; despite preprocessing efforts to reduce skewness and mitigate outliers, some distributional irregularities remained. Furthermore, the target variable contains a well-known capped value of 500001, which introduces additional distortion into the data distribution. These observations suggest that data quality, preprocessing, and feature engineering had a greater influence on the final performance.

2 | LOSS FUNCTION INSIGHTS

The results also reveal an interesting trend among the Adaptive Loss variants. As the parameter α moves toward more robust loss formulations, the R^2 score generally improves. This observation suggests that robust loss functions provide better stability and generalization when the dataset contains residual outliers, skewed distributions, or other non-ideal characteristics.



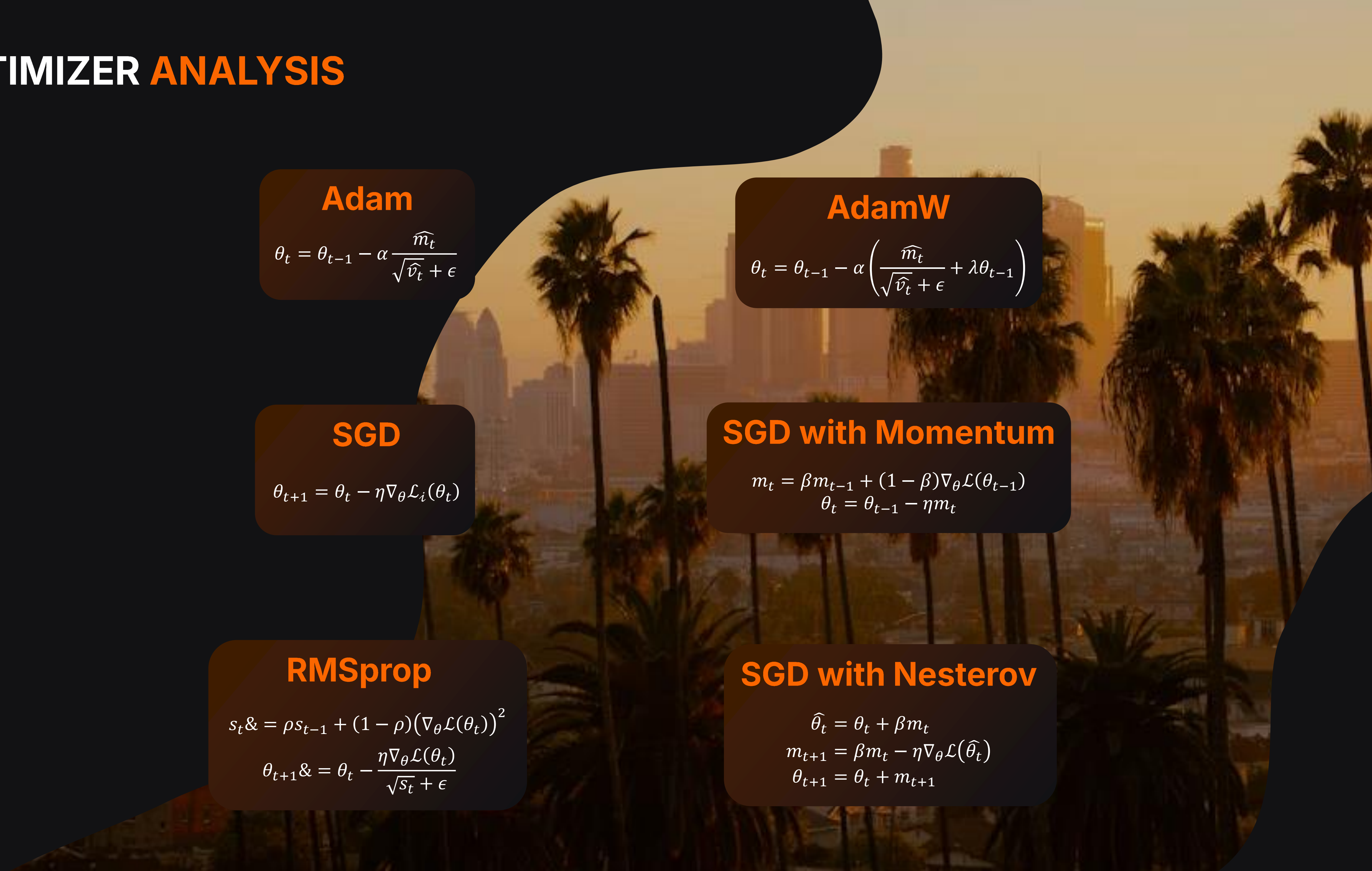
2 | LOSS FUNCTION INSIGHTS

Under these conditions, loss functions such as MSE and Quadratic Loss are highly sensitive to large prediction errors because they penalize errors quadratically. As a result, the model tends to focus excessively on a small number of samples with large residuals, such as very expensive houses or properties located in special districts. This may reduce the model's overall ability to generalize across the entire dataset.

In contrast, robust loss functions such as MAE and Welsch reduce the influence of extreme samples by limiting the impact of large residuals. Consequently, they achieve slightly better predictive performance on this dataset.

Overall, the experimental results indicate that robust loss functions consistently outperform traditional quadratic losses on the California Housing dataset, although the performance gap remains relatively small.

3 | OPTIMIZER ANALYSIS



Adam

$$\theta_t = \theta_{t-1} - \alpha \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}$$

AdamW

$$\theta_t = \theta_{t-1} - \alpha \left(\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right)$$

SGD

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}_i(\theta_t)$$

SGD with Momentum

$$m_t = \beta m_{t-1} + (1 - \beta) \nabla_\theta \mathcal{L}(\theta_{t-1})$$
$$\theta_t = \theta_{t-1} - \eta m_t$$

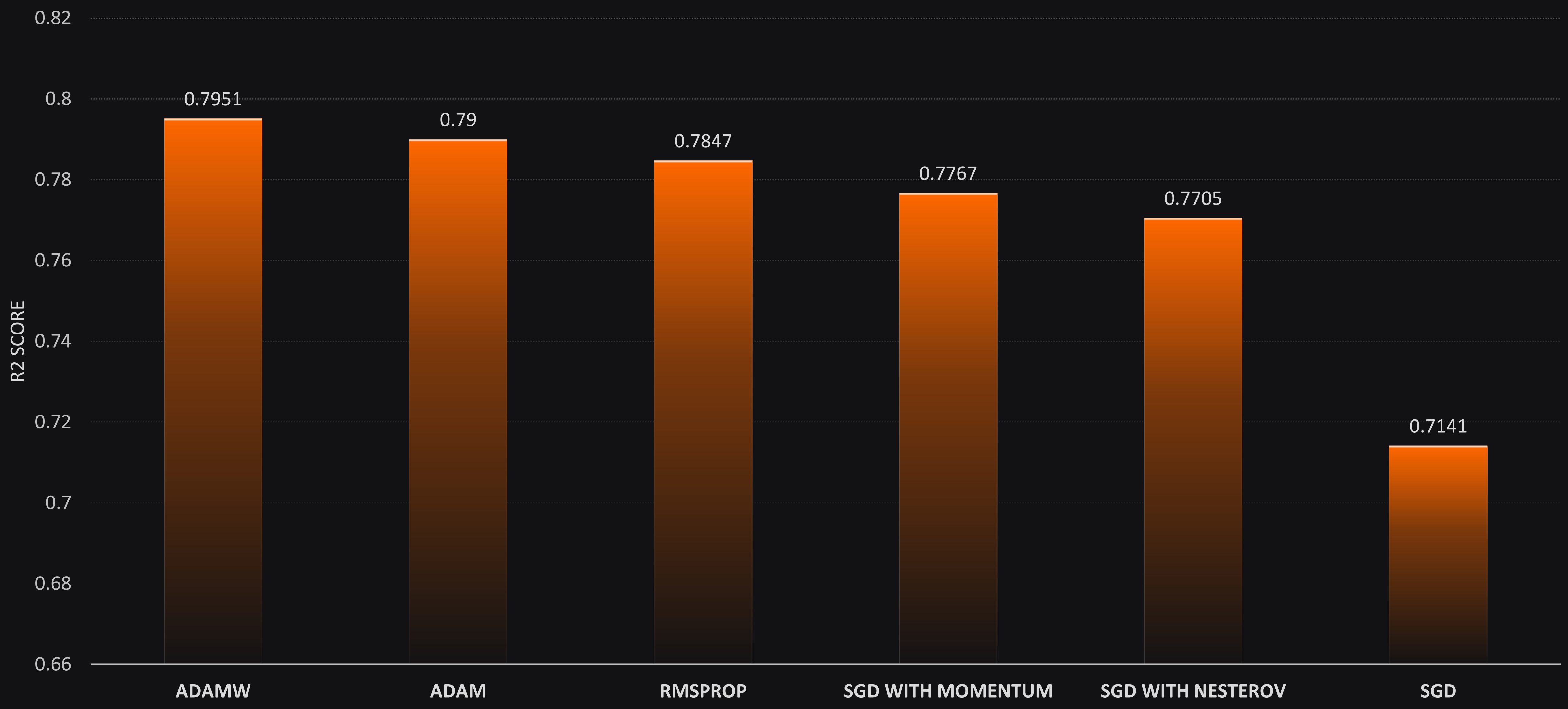
RMSprop

$$s_t \& = \rho s_{t-1} + (1 - \rho) \left(\nabla_\theta \mathcal{L}(\theta_t) \right)^2$$
$$\theta_{t+1} \& = \theta_t - \frac{\eta \nabla_\theta \mathcal{L}(\theta_t)}{\sqrt{s_t} + \epsilon}$$

SGD with Nesterov

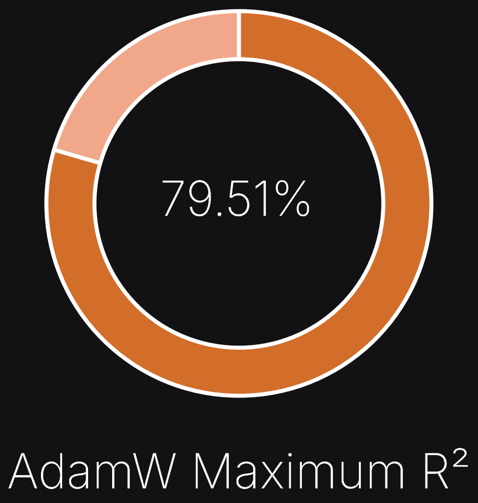
$$\widehat{\theta}_t = \theta_t + \beta m_t$$
$$m_{t+1} = \beta m_t - \eta \nabla_\theta \mathcal{L}(\widehat{\theta}_t)$$
$$\theta_{t+1} = \theta_t + m_{t+1}$$

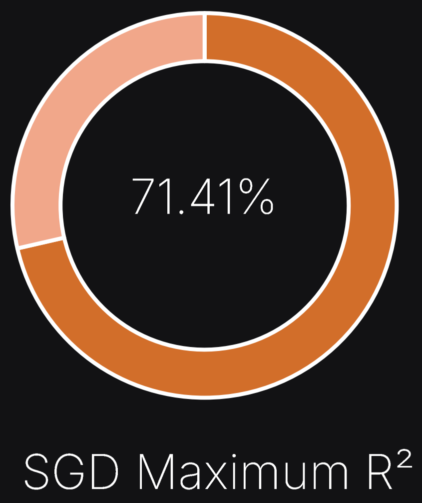
3 | OPTIMIZER RESULTS

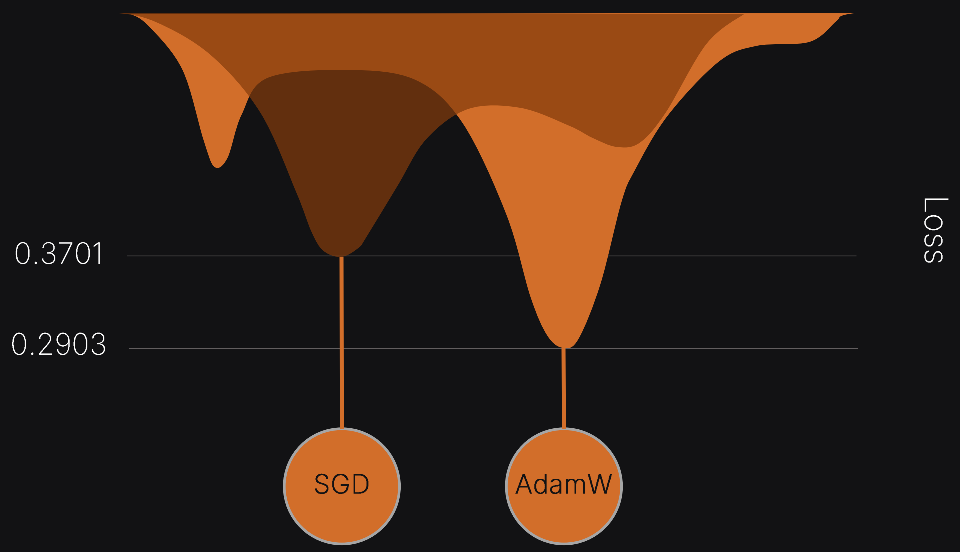


3 | OPTIMIZER INSIGHTS

The results indicate that AdamW achieved the highest validation performance with an R^2 score of 0.7951 and a loss of 0.3701, closely followed by Adam with 0.7900. The difference between these two optimizers is relatively small (0.0051), suggesting that both methods perform similarly on the California Housing dataset. RMSprop also produced competitive results with an R^2 score of 0.7847, remaining reasonably close to the two Adam-based optimizers. In contrast, SGD achieved a lower R^2 score of 0.7141, but with a smaller loss value of 0.2903, highlighting that loss values are not directly comparable across different optimization behaviors and should be interpreted alongside R^2.





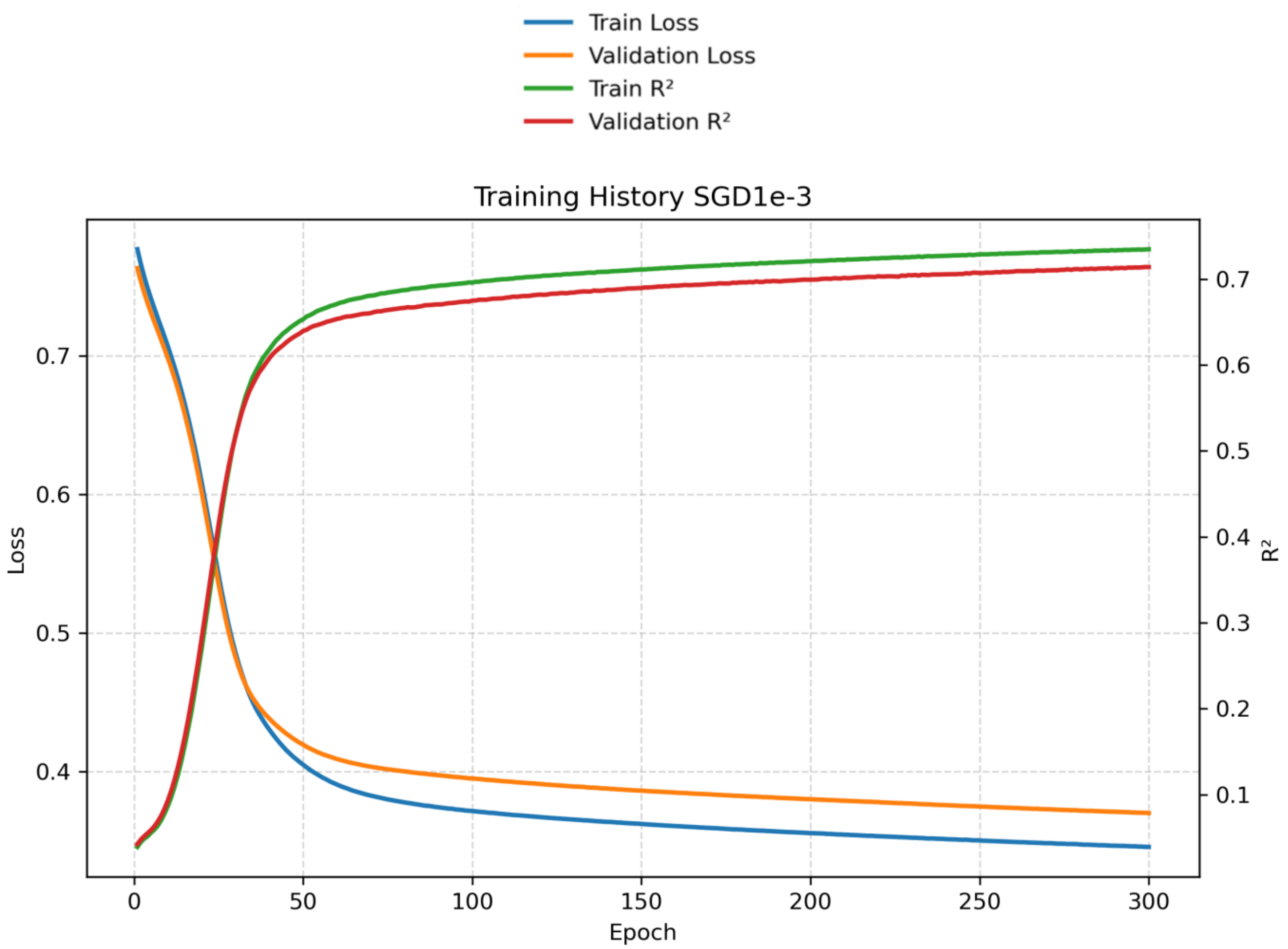



Training History AdamW
Loss
Epoch
R²
Train Loss
Validation Loss
Train R²
Validation R²
0.45
0.40
0.35
0.30
0.25
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0
50
100
150
200
250
300

3 | OPTIMIZER INSIGHTS

EFFECT OF LEARNING RATE

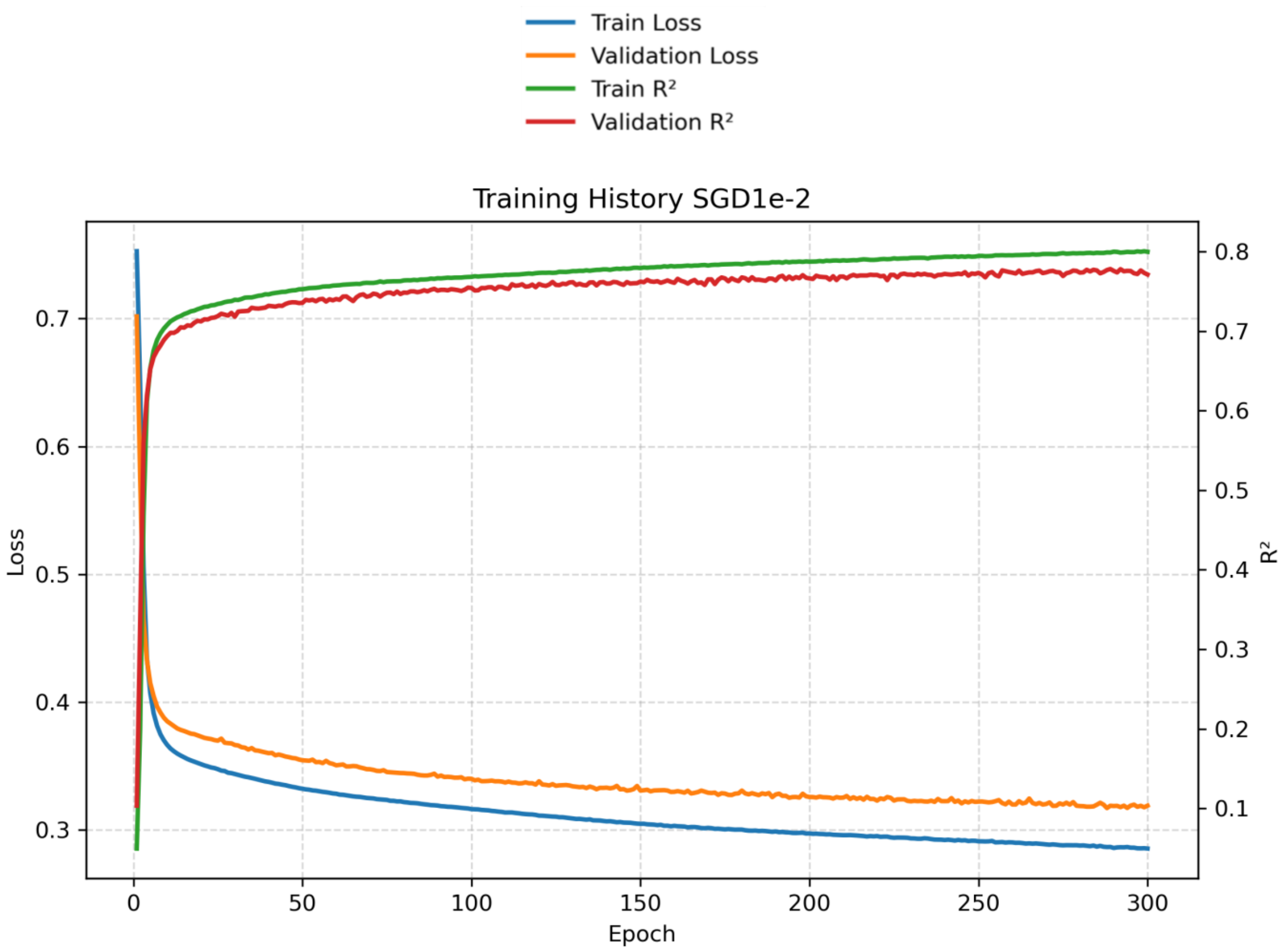
I analyzed SGD using three different learning rates (0.001, 0.01, and 0.1) to study their impact on convergence and model performance. With a learning rate of 0.001, the updates were very small, resulting in slow learning and clear underfitting. The learning curves were very smooth with almost no fluctuations, indicating that the optimizer was not making large enough steps to escape suboptimal regions.



3 | OPTIMIZER INSIGHTS

EFFECT OF LEARNING RATE

At a learning rate of 0.01, the model showed faster convergence and started to exhibit small oscillations in the loss curves, reflecting more aggressive parameter updates.



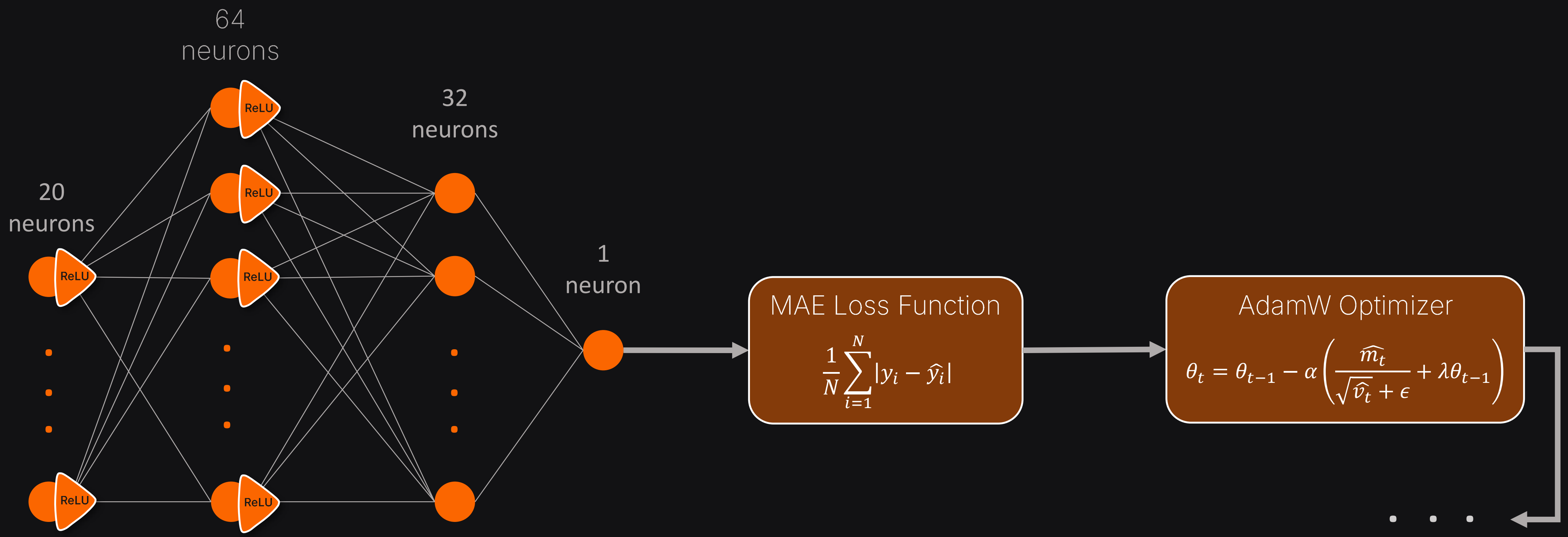
3 | OPTIMIZER INSIGHTS

EFFECT OF LEARNING RATE

With a learning rate of 0.1, the model achieved the best performance. However, the learning curves showed noticeable fluctuations due to large update steps, which sometimes caused overshooting before converging toward better solutions. Overall, the results demonstrate that the learning rate significantly affects both convergence behavior and predictive performance in SGD

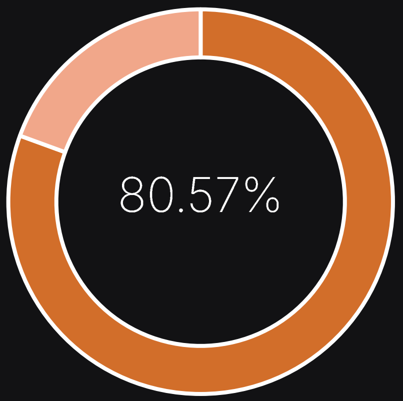
4 | FINAL TESTING

The final evaluation was performed using the best-performing model, which consists of three fully connected layers with ReLU activation functions, the MAE loss function, and the AdamW optimizer. The model was loaded from the saved checkpoint AdamW_best_model.pth.



4 | FINAL TESTING

The model achieves a strong generalization performance with an R^2 of 0.8057 on the test set, indicating that it effectively captures the underlying patterns of the California Housing dataset with stable and low prediction error.

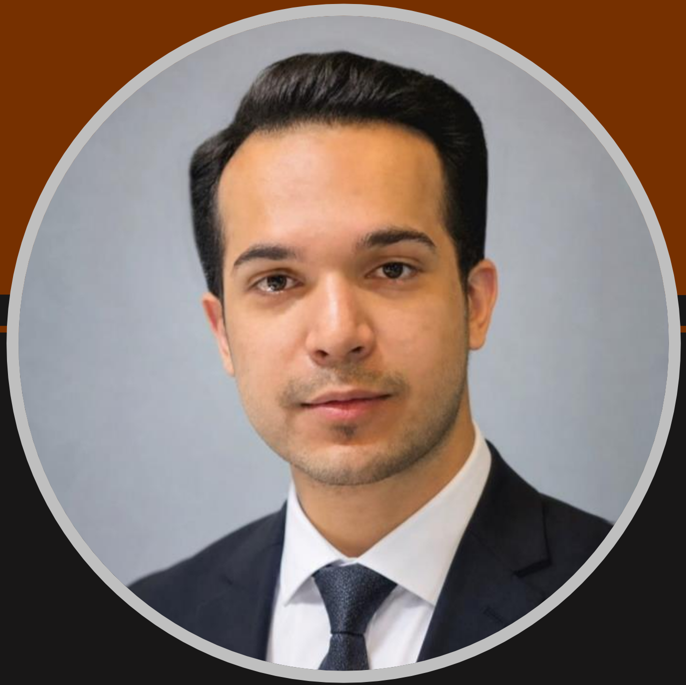


R^2 Score

FUTURE IMPROVEMENTS

Although the neural network achieved strong predictive performance on the California Housing dataset, it may not be the optimal model for this type of tabular data. Tree-based machine learning algorithms such as Random Forest, XGBoost, LightGBM, and CatBoost often outperform neural networks on structured datasets because they can naturally capture complex feature interactions, handle non-linear relationships, and are generally less sensitive to feature scaling and preprocessing. Exploring and comparing these models would be a valuable direction for future work and could potentially lead to improved predictive performance.

I welcome collaborations and contributions to further improve this project through the exploration of new models and techniques. You can reach out via email or LinkedIn to discuss potential improvements and future developments.



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Hi there! I'm a computer engineer passionate about Artificial Intelligence and Data Science. I truly believe this field is shaping our future, and I'm on a journey to continuously learn and make a meaningful impact. Feel free to follow my work if you're interested. I'm always excited to connect and collaborate with like-minded people in this space.